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RESEARCH ARTICLE

Hierarchical Multi-Stage Framework for Robust and Explainable Tomato Leaf Disease Identification

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ABSTRACT Accurate identification of tomato leaf diseases is essential for improving crop health management and reducing agricultural losses. However, the wide variety of tomato diseases, the limited availability of real-world datasets, and the variability of agricultural environments make developing reliable and explainable tomato disease identification systems challenging. Traditional deep learning models trained on controlled datasets often fail to generalize in real-world farm environments due to complex backgrounds and variations in illumination and image conditions. This paper presents a hierarchical, multi-stage framework that integrates leaf detection, segmentation, and classification with interpretability to achieve robust and explainable disease identification. The proposed pipeline employs YOLO11 for leaf detection, the Segment Anything Model (SAM) for segmentation, a ResNet-50 classifier, and LIME interpretability. To evaluate the impact of the pipeline on robustness and generalization, we conducted cross-domain experiments in two settings, using PlantVillage for laboratory conditions and PlantDoc for real-world conditions. The proposed pipeline improved real-world accuracy from 22.79 % with a flat ResNet-50 to 55.33 %, cutting the accuracy drop by 34.0 %. The pipeline stages focused on leaf isolation to ensure that predictions are driven by symptomatic tissue rather than background. Experiments demonstrate that our approach significantly improves generalization and transparency, addressing major gaps in existing plant disease identification systems.

INDEX TERMS Plant disease, YOLO11, SAM, tomato leaf detection, segmentation, ResNet-50, LIME, PlantVillage, PlantDoc.

I. INTRODUCTION

Tomatoes are one of the most grown crops worldwide, serving as a major source of income for millions of farmers [1], [2]. Tomatoes are also nutritious and make an essential contribution to healthy diets [3].

A major constraint in tomato production is the spread of fungal, bacterial, and viral diseases, such as Bacterial Spot, Late Blight, Leaf Mold, and Yellow Leaf Curl Virus [4].

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These diseases can spread rapidly, leading to serious yield losses and affecting both food security and agricultural profitability [5], [6]. Therefore, detecting such diseases early is an essential step toward sustainable and efficient crop management.

Traditional disease identification methods are based on manual field inspection, which requires expert knowledge, extensive labor, substantial time, and lacks the precision required for sustainable crop management [7]. In contrast, modern computer vision and deep learning techniques offer automated and more accurate alternatives to identify plant

diseases directly from leaf images [8]. Although these models perform well in controlled settings, their accuracy often degrades when applied to real-world agricultural scenes that include soil, stems, shadows, and noisy backgrounds. This domain gap between laboratory datasets and field images remains one of the main challenges in practical deployment [9], [10], [11]. Studies tend to report only training or validation results, as these usually show high accuracy under controlled conditions. However, real-world deployment of disease identification systems requires evaluating models on unseen field data with natural variations in lighting, background, and leaf appearance. Without such testing on real-world datasets, the reported performance can be misleading. A model that performs well during training may fail when exposed to a new environment. Furthermore, existing approaches cannot explain the behavior of the model, making it difficult to understand the factors that contribute to incorrect predictions and generalization gaps. Without interpretability, it becomes challenging to identify dataset biases, assess the model's reliance on irrelevant background features, or ensure that the learned representations correspond to true disease symptoms [12], [13], [14].

Addressing these problems requires not only powerful classifiers, but also a structured pipeline capable of isolating the relevant parts of the plant before analysis. In this work, we propose a hierarchical multi-stage framework that mirrors the logical sequence of human visual inspection, detecting the leaf, isolating it precisely, classifying the disease, and justifying the prediction. The proposed framework begins with leaf detection using the Ultralytics YOLO11 object detector to locate tomato leaves within complex backgrounds [15], [16]. The detected regions are then cropped and refined using the Segment Anything Model (SAM) to separate the leaf from surrounding elements and background noise [17]. The final disease classification is performed by a ResNet-50 convolutional neural network, and the reasoning behind each prediction is visualized using Local Interpretable Model Agnostic Explanations (LIME) to enhance transparency and explainability [18]. Because stages directly affect classification reliability, this work evaluates each stage of the pipeline separately and compares multiple leaf segmentation approaches using a manually labeled subset of the PlantVillage dataset.

The proposed integrated design significantly improves the robustness of disease classification systems in complex environments and makes predictions more interpretable for agricultural experts. Unlike conventional classifiers that perform single-stage classification, the hierarchical framework operates directly on the relevant plant tissue, minimizing bias from background regions and producing verifiable visual explanations of the predicted diseases. This work explicitly analyzes the generalization when dealing with real-world tomato images. We compare the performance of the pipeline under environment shifts, such as different lighting and backgrounds, against a flat CNN baseline to show that our multi-stage hierarchical approach can reduce

the performance drop under complex field conditions than a direct single-stage CNN classifier trained on data under lab conditions.

In summary, this paper presents a comprehensive and explainable approach to accurately identify tomato leaf disease in real-world conditions. The proposed approach combines modern object detection and segmentation techniques with interpretable deep learning in a multi-stage pipeline that focuses on the disease classification performance, generalization, and interoperability. The resulting system represents a step toward reliable and transparent plant disease diagnosis systems.

The remainder of this paper is organized as follows. Section II reviews the recent literature on the detection, segmentation, and classification of tomato leaf diseases. Section III describes the proposed hierarchical multi-stage pipeline, including datasets description, detection, segmentation, classification, and interpretability methods, as well as the evaluation processes. Section IV presents the experimental results with a discussion on the performance and comparative analysis of the proposed pipeline. Finally, Section V concludes the paper.

II. RELATED WORK

Artificial Intelligence (AI) and computer vision (CV) techniques have significantly advanced agriculture in the last decade [19], [20]. They have provided solutions to agricultural problems by supporting the application of precision farming methods and enhancing sustainable agriculture systems. Automating the detection and classification of plant diseases from leaf images has become one of the most important and attractive tasks in this field [21]. Tomato, one of the most widely cultivated and consumed crops, has been the focus of many studies that aim to improve disease identification accuracy and field-level deployability. This section reviews key contributions related to tomato leaf identification and positions the proposed hierarchical pipeline within this research context.

A. CONVENTIONAL TOMATO LEAF DISEASE IDENTIFICATION METHODS AND GENERALIZATION ISSUES

Existing single-stage approaches typically train a single deep learning model for disease detection, classification, or segmentation using different labeled disease datasets. A previous review [22] presented an extensive survey of deep learning applications for the detection, segmentation, and classification of tomato leaf disease between 2019 and 2023. Their review showed that CNN-based architectures such as VGG-16, ResNet-50, InceptionV3, EfficientNet, and XceptionNet achieved accuracies greater than 98% on the PlantVillage dataset. Similarly, a comparative analysis on the PlantVillage dataset reported accuracies of 98.60%, 92.39%, 96.15%, and 98.98% for CNN, VGG-16, VGG-19, and ResNet-50 models, respectively [23]. ResNet-50 outperformed other architectures due to its residual connections

that mitigate gradient vanishing and improve feature reuse, making it very efficient for plant disease classification tasks.

However, these models were trained and tested on homogeneous datasets with uniform lighting and backgrounds, resulting in limited robustness and reduced accuracy when applied to complex field imagery, such as the PlantDoc dataset. A network trained on PlantVillage in lab settings achieved as low as 13–33% accuracy on the PlantDoc field dataset, indicating poor real-world efficacy without domain adaptation [9]. One of the attempts to address this issue is MLFAnet [24], which proposed a tomato disease classification network designed to measure and improve generalization under out-of-distribution (OOD) corruption shifts. Unlike many studies that test on data drawn from the same distribution as training, MLFAnet creates a corruption-shift leaf testing set (CLT) based on ImageNet-C corruptions. Similarly, a previous study [25] employed a Vision Transformer integrated with a mixture of experts approach and evaluated the domain gap between the in-domain and out-of-domain settings. The accuracy of the model decreased from 99.96% to 74% using an out-of-domain subset of the PlantDoc dataset; however, this subset comprised only 24 images, limiting the reliability of the reported generalization performance. Another work [26] observed a drop from 99.84% on PlantVillage to 41.81% on their Field-PV images. They reduced this gap by designing an improved CNN, which increased field accuracy to 72.03%.

Isolated approaches for the identification of leaf diseases often result in suboptimal performance in challenging conditions, such as different leaf shapes and complex environments. Thus, combining multiple approaches improves generalization and robustness. A prior work [27] presented multiple pre-processing stages to extract leaves before classification using canny edges and intensity of color. By highlighting edges and segmenting the leaf, canny edge detection provided isolated leaf images to the training process. The combined approach used provided a more robust disease identification system with in-domain accuracies of 99.03 % and 98.23 % achieved by the DenseNet201 and EfficientNetB3 models, indicating the need for leaf isolation to achieve accurate disease classification results. However, leaf isolation using colors and edges performs poorly when backgrounds contain noise and objects with similar textures or colors.

B. OBJECT DETECTION LEAF LOCALIZATION APPROACHES

Object detection techniques have recently been used to provide more accurate object localization in different environments [28]. Object detection frameworks have evolved from computationally heavy two-stage detectors to real-time single-stage detectors such as YOLO (You Only Look Once). Two-stage architectures typically offer higher precision, while single-stage models perform detection in a single forward pass, achieving high-speed advantages. According to a previous research [29], Fast R-CNN achieves around

70% detection accuracy, while YOLO reaches 63.4%, but with inference speeds approximately 300 times faster. Subsequent YOLO versions introduced anchor-free prediction, attention modules, and multi-scale processing, achieving accuracies competitive with two-stage architectures. These characteristics make YOLO models especially suitable for real-time agricultural applications, including fruit counting, weed detection, and disease localization under diverse field conditions. In a previous study [30], YOLO11 was trained using the AgriVision-Kiwi dataset to detect and isolate leaf regions before moving to recognition of kiwi leaf diseases. Accurate leaf detection minimized background noise and improved the reliability of subsequent classification tasks.

C. SEGMENTATION FOR LEAF ISOLATION

In the context of tomato disease identification, segmentation is typically used to identify and segment disease symptoms in tomato leaves. However, few studies have discussed the importance of using leaf segmentation as a preprocessing stage to isolate regions of interest (ROI) and eliminate background effects on classification performance and generalization. An ensemble-learning study utilized PlantVillage and Taiwan Tomato Leaf datasets to classify 13 classes of tomato diseases [31]. Multiple preprocessing pipelines, feature extraction techniques and 260 ensemble classifiers were evaluated, achieving a peak accuracy of 95.98%. The study highlighted the importance of segmentation to address variability caused by lighting changes, shadowing, and multiple overlapping leaves. Another work proposed a segmentation-driven pipeline based on a Multivariate Normal Deep Learning Neural Network (MNDLNN) [32]. In that framework, the images were converted into the HSI color space and the green pixels were masked to extract leaf regions. Unhealthy regions were segmented using rectilinear K-means clustering (RKMC). The study demonstrated that effective segmentation and feature space optimization significantly improved the accuracy of disease classification, even under challenging visual conditions. Further work extended this idea to video-based disease detection using deep ResNet architectures [33]. In that study, the tomato leaf regions in the field videos were extracted using simple RGB color-based object localization and processed using transfer learning with pre-trained ResNet101 and ResNet152 models. Although the previous approaches achieved strong performance, they relied on color-based segmentation and lacked automated leaf isolation.

Recent advances in foundation models have introduced a great shift in image segmentation. The Segment Anything Model (SAM), developed by Meta AI, enables zero-shot segmentation without using previously labeled training data. SAM can operate through user prompts (points, boxes, polygons) or fully automatically, creating masks for all objects in an image. The Leaf-Only SAM framework applied SAM with custom post-processing steps for potato leaf segmentation [34]. These steps included filtering non-green

pixels, eliminating multiple leaf masks, and refining shapes to retain single leaf instances. Although **Leaf-Only SAM achieved slightly lower precision and recall (73.9 % and 73.1 %) compared to a fine-tuned Mask R-CNN (84.4 % and 87.9 %)**, it did not require annotated data, provided automatic leaf isolation, and generalized well across unseen tomato leaf datasets.

Other works integrated detection and segmentation in a unified pipeline. An attempt [35] used a YOLOv8-seg model for automated leaf instance segmentation and then improved it by introducing a Ghost module and a Bidirectional Feature Pyramid Network (BiFPN) into the standard architecture to better capture subtle leaf boundaries and multi-scale features. Another previous study [36] presented an automated framework that uses SAM for semantic segmentation and YOLOv8 for real-time detection. SAM generates leaf masks that are converted to YOLOv8 bounding boxes, providing automatic image labeling. Other researchers have integrated YOLOv8 and SAM to prepare images of tomato leaves for further analysis, such as real-time disease classification [37]. Their work concluded that leaf detection followed by leaf segmentation can provide clean and isolated leaf images that can be used as input to other tasks. In addition, it proved that SAM performed well for prompt-based segmentation.

D. EXPLAINABILITY IN PLANT DISEASE DIAGNOSIS MODELS

Explainable AI (XAI) has recently received significant attention due to its ability to make learning models more transparent, interpretable, and trustworthy by allowing humans to understand the reasoning behind their predictions [38]. In agriculture, explainability provides valuable information to help farmers and experts visualize, understand, and confirm predictions [39], [40]. In a related study [41], **EfficientNetB5 was used to classify nine classes of tomato diseases in addition to healthy leaves, integrating Grad-CAM and LIME to provide interpretability, thus visualizing which image regions had the most influence on classification decisions.** Another work [42] proposed XSE-TomatoNet, which enhances EfficientNetB0 with Squeeze-and-Excitation blocks and multiscale feature fusion enhanced with LIME, SHAP, Grad-CAM, and Grad-CAM++ for interpretability, illustrating how explainability techniques can be integrated into high-performance disease classifiers.

E. SUMMARY AND RESEARCH GAP

Existing studies have advanced automated plant disease diagnosis using deep learning, particularly in detection, segmentation, and classification. However, **most single-stage models are trained and evaluated on the same dataset,** achieving high in-domain accuracy, but poor generalization when tested in the field or on unseen conditions. Most existing approaches do not analyze the generalization ability of their classification models in the wild, and the studies that evaluated generalization typically added noise to the

same dataset rather than testing on a completely different real-world dataset. Explainable AI techniques such as LIME and Grad-CAM improve model transparency; however, they are rarely utilized as evaluation tools to determine the limitations of existing models and approaches, as well as to visually evaluate how the background components affect generalization.

Although several works have focused on improving network architectures, they often overlook the influence of preprocessing on the final classification performance and results. Single-stage classification methods without isolating leaves depend on background components, particularly under complex real-world conditions. Object detection techniques have been used to detect diseased regions, but they often function as independent modules rather than preprocessing stages that prepare data for further phases, such as classification. Segmentation methods for leaf isolation have demonstrated their positive impact on classification accuracy by reducing background noise, but they still lack the automated and reliable integration to ensure consistent performance in different environmental conditions. Although few works have applied leaf detection and segmentation, they relied on edge and color methods, making them ineffective when backgrounds are sharp, contain visible stems, or vary in brightness. These approaches also fail when leaf diseases cause changes in color or texture.

Despite these developments in advancing detection, segmentation, classification, or interpretability, **none combine them into a unified, end-to-end pipeline that aims to improve generalization and increase disease prediction performance under complex field conditions rather than controlled laboratory settings.** The characteristics of different works on the classification of tomato diseases are summarized in Table 1, highlighting the generalization and interpretability gaps addressed in this work.

TABLE 1. Summary of tomato disease classification studies. Gen.=generalization; Interp.=interpretability; Det.=detection; Seg.=segmentation. “-” = not reported.

Ref	Model	Gen.	Interp.	Det.	Seg.
[31]	260 Ensembles	-	-	×	✓
[32]	MNDLNN	-	-	×	✓
[33]	ResNet101/152	-	-	×	✓
[23]	CNN/VGG/ResNet	×	-	×	×
[41]	EfficientNetB5	-	✓	×	×
[42]	XSE-TomatoNet	-	✓	×	×
[24]	MLFAnet	✓	-	×	×
[25]	ViT+MoE	✓	×	×	×
[9]	GoogLeNet/ResNet18	✓	×	×	×
Ours	ResNet-50	✓	✓	✓	✓

III. METHODOLOGY

This section describes the proposed pipeline and the materials used in this study, including details on the datasets, the proposed stages, evaluation approaches, and metrics, and the implementation setup used for experimentation.

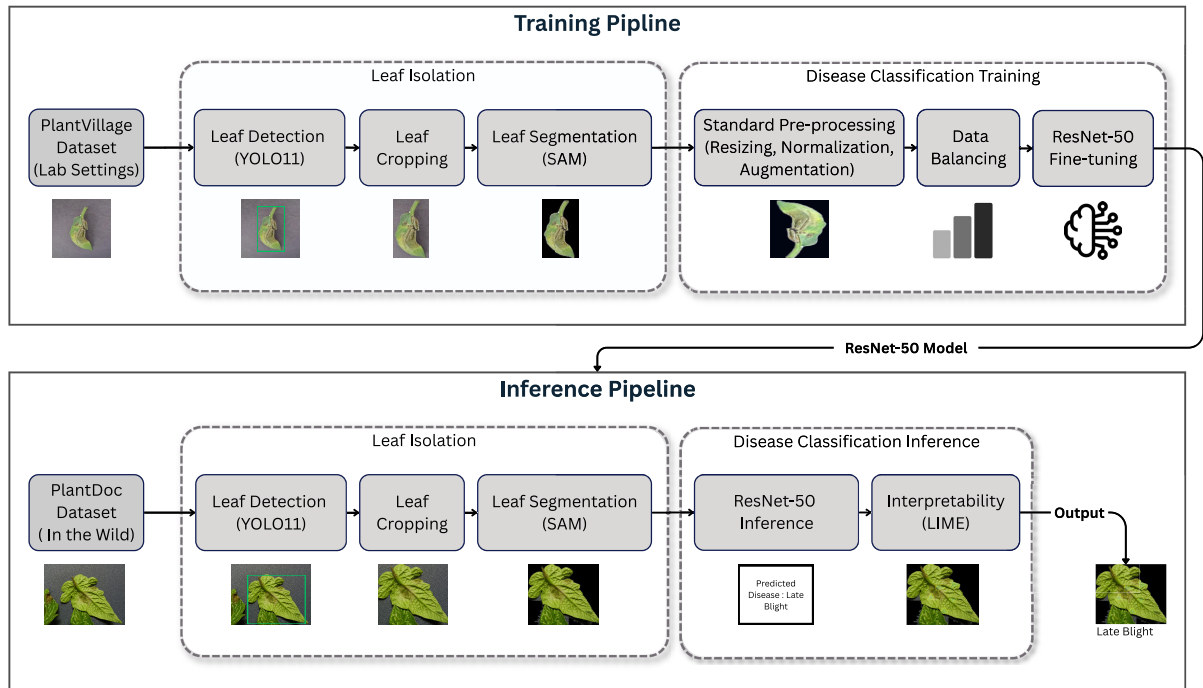


FIGURE 1. An overview of the proposed framework.

A. OVERVIEW OF THE PROPOSED METHOD

The proposed framework, illustrated in Fig. 1, presents a hierarchical deep learning pipeline for accurate, robust, and explainable identification of tomato leaf disease. The workflow is divided into two main parts, a training pipeline and an inference pipeline. In the training pipeline, the PlantVillage dataset is processed through a sequence of leaf isolation units, including detection using YOLO11, cropping to extract regions of interest, and segmentation using the Segment Anything Model (SAM) followed by preprocessing and disease classification training using a ResNet-50 network. In the inference pipeline, leaf images from the real-world PlantDoc dataset undergo the same leaf isolation process, after which the trained model performs disease classification, interpretability analysis using LIME, and generalization evaluation.

B. DATASET DESCRIPTION

This section describes the datasets used at the different stages of the pipeline to train and evaluate the proposed system. Each dataset serves a specific purpose within the workflow, supporting model learning under controlled conditions, examining generalization to real-world environments, and verifying the robustness of the overall system. Detailed information on the composition of the datasets, image characteristics, and usage context is provided in the following.

1) PlantVillage DATASET

PlantVillage [43] is one of the most popular and comprehensive open source agricultural image datasets and is used

TABLE 2. Tomato leaf disease classes, abbreviations, and brief symptoms.

Code	Disease Name	Symptoms
YLCV	Yellow Leaf Curl Virus	Curling, yellowing, stunting.
LB	Late Blight	Brown spots, white mold.
MD	Mosaic Disease	Mottled color, distortion.
BS	Bacterial Spot	Dark spots, yellow halo.
SM	Spider Mite	Yellow dots, dry leaves.
SLS	Septoria Leaf Spot	Gray spots, dark edge.
LM	Leaf Mold	Yellow spots, gray mold.
TS	Target Spot	Brown rings, yellow edge.
H	Healthy	Normal green leaves.

as a benchmark in leaf disease identification research [44]. It contains high-quality, laboratory-captured images (256 × 265, RGB) of healthy and diseased plant leaves under controlled lighting and uniform backgrounds. For this study, a subset of the PlantVillage dataset was used, covering nine classes of tomato leaf diseases, including Bacterial spot (BS), late blight (LB), leaf mold (LM), mosaic disease (MD), yellow leaf curl virus (YLCV), septoria leaf spot (SLS), target spot (TS), spider mite (SM), and healthy (H). Each disease presents visually distinct features such as spot color, shape, and distribution as shown in Table 2. The PlantVillage dataset served as a training source for the classification stage of the proposed pipeline and was used for model validation and in-domain evaluation.

2) PlantDoc DATASET

Unlike PlantVillage, PlantDoc [9] provides in-the-wild plant disease images with image-level labels and substantial

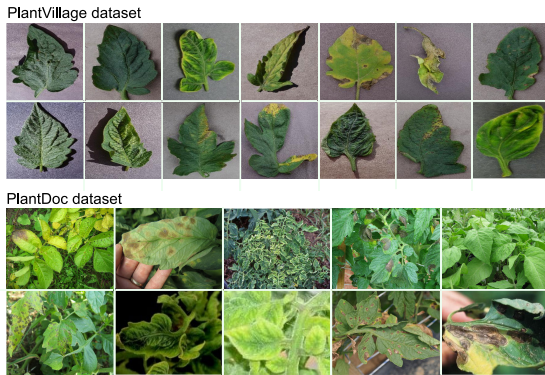


FIGURE 2. Samples from PlantVillage dataset (lab settings) and PlantDoc dataset (in the wild).

background, illumination, and viewpoint variability, allowing for evaluation under realistic field conditions. It was collected from the publicly available web and field photographs contributed by various agricultural research sources and repositories. A group of sample images from this dataset is shown in Fig. 2 compared to samples from the PlantVillage dataset, showing the different conditions between them. This domain shift introduces significant challenges for visual learning models, making it ideal for evaluating the generalization and robustness of trained models. **The tomato classes from the PlantDoc dataset were used for real-world evaluation to evaluate domain generalization and performance, and were not used for training at any stage.** The subset used in this study contained six classes selected to overlap with the tomato classes in the PlantVillage dataset, with all available images from these classes included in the evaluation.

3) ROBOFLOW LEAF DETECTION DATASET

To train the YOLO11 model for automatic leaf detection, **an open-source Roboflow dataset containing 2,340 images** of plant leaves was used. The dataset contains images of multiple plant species captured under diverse environmental conditions, such as variations in illumination, background complexity, and leaf appearance and orientation. In the dataset, images are annotated with bounding boxes that precisely outline individual leaves, providing a reliable source for object detection training. The diversity of the dataset allows the YOLO11 detector to accurately locate leaves in both controlled laboratory environments and field conditions where leaf images vary significantly.

C. LEAF DETECTION

The first stage of the proposed pipeline involves automatic leaf detection and cropping using the YOLO11 object detection model. This stage is responsible for localizing individual leaves within each image and preparing region-of-interest (ROI) inputs for subsequent segmentation and

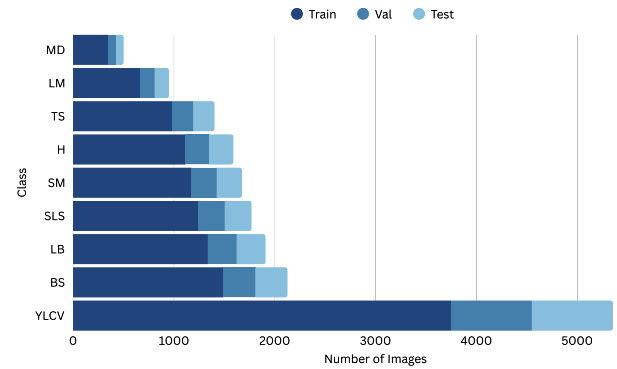


FIGURE 3. Dataset splits, training, validation, and testing.

disease classification stages of the pipeline. YOLO11 was chosen due to its strong performance in real-time object detection and its ability to maintain high precision under varied environmental conditions. It offers an optimal balance between speed, accuracy, and generalization for small-object detection tasks such as plant leaf detection.

The model was trained within the Ultralytics YOLO11 framework, initialized with the pretrained YOLO weights (yolo11n.pt) for transfer learning. The training used the default configuration with an image size of 640×640 pixels, a batch size of 16, and the Stochastic Gradient Descent (SGD) optimizer with a momentum of 0.937, weight decay of 0.0005, and an initial learning rate of 0.01 following a cosine decay schedule. **Training was executed for 50 epoches** to fine-tune the model on the previously described Roboflow leaf dataset, that was split into 80% training and 20% validation sets.

During training, the model learned to detect leaf regions, performing robust and accurate leaf extraction. After training, the YOLO11 detector was integrated as the entry point of the pipeline to automatically identify leaf areas and crop them as regions of interest from raw images. This step ensures that SAM segmentation and ResNet-50 disease classification modules receive clean leaf input images, improving overall pipeline accuracy and generalization efficiency. The bounding boxes generated at this stage were cropped and passed to the segmentation module.

D. LEAF SEGMENTATION

Following the detection and cropping of leaf regions using YOLO11, the next stage in the proposed pipeline performs leaf segmentation using the Segment Anything Model (SAM) to isolate the leaf surface and remove background components such as stems, soil, or shadows. Accurate segmentation is essential to ensure that the following disease classification focuses exclusively on the leaf, which is the region of interest (ROI) in this system. **Each leaf cropped by YOLO11 was passed to SAM, which automatically generated multiple segmentation masks.** The largest connected mask in the YOLO11 cropped images was selected as the main

leaf area, while smaller masks were filtered out. The mask was applied using bitwise operations to remove noisy backgrounds, producing clean isolated leaf regions. The produced background-free leaf images serve as refined inputs to the ResNet-50 classification network, ensuring that the disease classification stage processes only precise representations of each leaf.

E. DISEASE CLASSIFICATION USING ResNet-50

After leaf detection and segmentation using YOLO11 and SAM, the resulting background-free leaf images are fed into a deep convolutional network for disease classification. In what follows, we describe the data preprocessing and classification training procedures.

1) DATASET SPLITTING AND PREPROCESSING

The dataset was partitioned into training, validation, and test sets with a split of 70%/15%/15% per class as shown in Fig. 3. It also shows the number of images per class used from the PlantVillage dataset. The training set was used to train the classification model, and the validation set was used to validate the performance of the model and tune the hyperparameter.

The preprocessing stage was designed to prepare and form the input images for effective training, improved model generalization, and reliable results. Each image was first resized to a fixed resolution of 64×64 pixels to ensure consistency throughout the dataset. The classification model operates on cropped leaf regions, which preserves the relevant visual cues and allows effective classification at this input resolution. The pixel values were then normalized to a range of $[0, 1]$ to accelerate convergence and lower the training error. During data augmentation, techniques, including random flipping, rotation, and brightness changes, were applied to the training set to improve robustness and reduce overfitting. Finally, the Weighted Random Sampler sampling strategy was employed to address class imbalance, ensuring that all disease categories contributed equally during training and model optimization.

2) MODEL ARCHITECTURE AND TRAINING SETUP

A customized ResNet-50 architecture was used, configured for nine output classes corresponding to the selected tomato disease categories in the PlantVillage dataset. The model was optimized using the AdamW optimizer with a learning rate of 3×10^{-4} , a weight decay of 1×10^{-4} , and a cosine annealing learning rate schedule following an initial warm-up phase. The training process was for 50 epochs with a batch size of 64 for both training and validation, and 16 for testing. The model checkpoints were saved according to the best validation accuracy. The hyperparameter selection method began by using the default parameters to train the model and then manually tuning them to find the best set of hyperparameters. Table 3 summarizes the hyperparameters used in training the ResNet classifier.

TABLE 3. Training Hyperparameters for the ResNet-50 Classifier.

Parameter	Value
Input size	64×64 RGB
Batch size	64 (train/val), 16 (test)
Learning rate	3×10^{-4}
Weight decay	1×10^{-4}
Epochs	50
Scheduler	Warm-up + Cosine Decay

F. MODEL INTERPRETABILITY WITH LIME

Local Interpretable Model-Agnostic Explanations (LIME) was used at inference for two aims. The first is to provide human-interpretable evidence of the regions driving each prediction and the second is to examine the influence of background noise on disease classification. LIME takes the trained model and an image instance, perturbs the image, observes modifications in the prediction, and identifies the regions that most influence the model's decision. Therefore, LIME highlights the image regions that correspond to visible disease symptoms in tomato leaves. This supports the feedback loop within the system, enabling continuous enhancement and performance improvement. LIME begins by segmenting each input image into superpixels using the Quickshift algorithm and then generates a set of perturbed samples by randomly masking combinations of these superpixels. Each perturbed image is passed to the trained ResNet-50 model that returns the corresponding class probabilities. Subsequently, LIME fits a locally weighted, L1-regularized linear model around the original image to approximate how each superpixel influences the class probability. This surrogate model captures the local decision boundary of the deep network, allowing estimation of the importance of each superpixel. For visualization, the predicted class was extracted and the K regions that contributed the most positively were identified. The resulting binary mask highlights the superpixels that most increased the predicted class probability. A light-green overlay was applied to these regions, and edges were emphasized using marking boundaries to provide clearer visual interpretation. This explanation reveals which parts of the leaf image most strongly supported the classification model's decision, offering an interpretable mapping between visual disease symptoms and the model's output.

G. PER-STAGE ASSESSMENT AND PERFORMANCE METRICS

The overall performance of the proposed pipeline depends on the accuracy, robustness, and reliability of each processing stage. This section includes the approaches and metrics used to examine how each stage, including detection, segmentation, and classification, achieves strong and reliable performance, ensuring consistent effects throughout the pipeline.

Starting with leaf detection, the training performance metrics of the YOLO11 model were calculated as in (1), (2), and (3), representing precision, recall, and

F1-score, respectively. Detection performance was further evaluated using mean Average Precision (mAP), reported as mAP@0.5 and mAP@0.5:0.95. After training, the performance of YOLO11 was tested and evaluated on images from both PlantVillage and PlantDoc datasets to ensure its robustness under different conditions and environmental settings.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (1)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (2)$$

$$F_1\text{-score} = \frac{2 \times TP}{2 \times TP + FP + FN} \quad (3)$$

where TP , FP , and FN denote true positives, false positives, and false negatives, respectively.

To determine the most effective segmentation approach for the proposed framework, three methods were examined and compared using a manually labeled subset of images taken from the PlantVillage dataset.

1) MARKER-BASED SEGMENTATION

This approach, previously proposed in [45], implemented morphological operations combined with the watershed algorithm to separate foreground and background regions of the image. The process begins with background marker creation, where color, texture, and intensity features identify non-leaf areas. Then, leaf markers are generated using grayscale morphological and edge-based operations to locate the leaf interior precisely. These two markers are passed to the segmentation process that divides the image into leaf and non-leaf regions. After segmentation, post-processing includes refinement using vein detection and contour smoothing to remove irregular edges in the resulting segmented image.

2) U-NET SEGMENTATION

A convolutional encoder-decoder network based on the ResNet-U-Net architecture was used for the semantic segmentation of the tomato leaf regions. The model integrates a ResNet backbone with a U-Net decoder to combine high-level contextual features and spatial details through skip connections. The U-Net architecture was implemented using the open source leaf segmentation U-Net framework [46].

3) SEGMENT ANYTHING MODEL (SAM)

SAM is a foundation-scale Vision Transformer (ViT) developed by Meta AI, trained on 11 million images and 1.1 billion masks. SAM is automatic and prompt-free, enabling zero-shot segmentation without the need for additional fine-tuning. It generalizes very well across different lighting, color, and background settings. To evaluate the performance of SAM, a ground truth subset of 50 images from the PlantVillage dataset was manually prepared and annotated using Label Studio. The images were equally distributed among all classes of tomato disease. This process is presented in Fig. 4 on a sample leaf image from the selected subset,

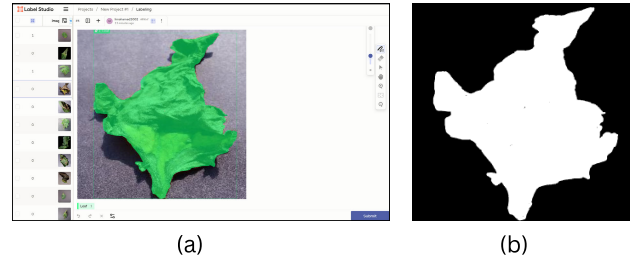


FIGURE 4. Ground truth generation process (a) Label Studio annotation for a sample leaf image. (b) The generated output mask.

showing the annotation output as an exported mask. The leaves were accurately annotated to provide a clear ground truth for a subset of the PlantVilge dataset because the open-source available PlantVillage segmented datasets are inaccurate, especially in images where the leaves show severe symptoms. These annotations were used to quantitatively evaluate segmentation performance for the three approaches using the standard metrics in (4), (5), and (6). Among the compared methods, SAM was selected for the segmentation task in the proposed pipeline.

$$IoU = \frac{TP}{TP + FP + FN} \quad (4)$$

$$Dice = \frac{2 \times TP}{2 \times TP + FP + FN} \quad (5)$$

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (6)$$

where TP , FP , and FN represent true positives, false positives, and false negatives, respectively.

The classification model was trained and validated on the PlantVillage dataset training, validation, and testing splits after passing the images through the proposed detection and segmentation stages. The training process was optimized by minimizing the cross-entropy loss shown in (7) and the classification performance was evaluated using accuracy as shown in (8). Generalization was evaluated by testing on PlantDoc, which contains field images with variable lighting and background conditions, to evaluate the performance of the model in the field.

$$\mathcal{L}_{CE} = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C y_{i,c} \log(\hat{y}_{i,c}) \quad (7)$$

where $\mathcal{L}_{CE}$ is the cross-entropy loss, N is the batch size, and $y_{i,c} \in \{0, 1\}$ is the one-hot encoded ground-truth label.

$$\text{Acc} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}(\arg \max_c \hat{y}_{i,c} = y_i) \quad (8)$$

where Acc denotes classification accuracy, y_i is the true class index, and $\mathbf{1}(\cdot)$ is the indicator function that equals 1 when the condition holds and 0 otherwise.

For comparison, a flat CNN baseline was trained on the same PlantVillage training images without passing through

the pipeline. The baseline used the same preprocessing techniques, data splits, optimizer, learning rate schedule, batch size, and number of epochs to ensure a fair comparison.

H. GENERALIZATION STUDY DESIGN

To assess the robustness of the model beyond the in-domain data, a cross-domain generalization test was conducted. The two classifiers, the flat and the pipelined ResNet, were evaluated under identical settings on the laboratory PlantVillage set and the real-world PlantDoc set. For each model, the accuracy and macro-F1 values were reported for tests on both datasets. The domain shift was calculated as the absolute and relative drops from laboratory to real-world settings (Accuracy ↓, Macro-F1 ↓). This approach identifies which model generalizes better to natural, field images and quantifies robustness to background and illumination variability. The real-world evaluation implemented aimed to highlight the generalization advantage of the proposed hierarchical pipeline over a flat ResNet baseline. The drop definitions used in this evaluation process were calculated using relative and absolute drops. Fig. 5 represents the generalization analysis procedure used to measure the change in pipeline performance when used in real-world images and compare it with the baseline performance.

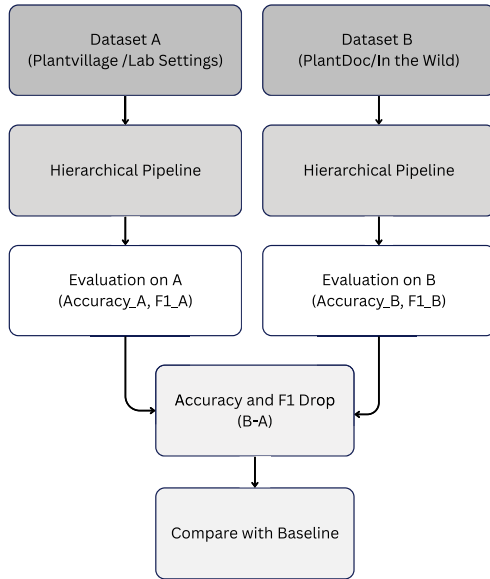


FIGURE 5. Generalization analysis procedure.

Let Acc_{in} and Acc_{ood} denote accuracy in-domain and out-of-domain, and similarly $F1_{in}$ and $F1_{ood}$ for macro-F1. We report both absolute drops in percentage points (pp) and relative drops:

$$\Delta_{Acc}^{abs} [pp] = Acc_{in} - Acc_{ood}, \quad (9)$$

$$\Delta_{Acc}^{rel} [\%] = \frac{Acc_{in} - Acc_{ood}}{Acc_{in}} \times 100, \quad (10)$$

$$\Delta_{F1}^{abs} [pp] = F1_{in} - F1_{ood}, \quad (11)$$

$$\Delta_{F1}^{rel} [\%] = \frac{F1_{in} - F1_{ood}}{F1_{in}} \times 100. \quad (12)$$

LIME was run on both classifiers, the flat ResNet-50 and the pipelined ResNet-50, to enable comparison under varying backgrounds and lighting conditions. The LIME resulting attribution maps were inspected to determine whether predictions were supported by true symptom regions on the leaf surface or by non-leaf background elements, revealing differences in model behavior, robustness, and generalization ability across the two approaches.

IV. RESULTS AND DISCUSSION

This section presents and discusses the results of the quantitative and qualitative evaluation of the proposed hierarchical pipeline for the identification of tomato leaf disease. First, the results of the stage-level evaluation that was performed on both the controlled PlantVillage dataset and the real-world PlantDoc dataset to measure detection, segmentation and classification performance, as well as interpretability. Second, the results of the cross-domain evaluation that was conducted to measure the generalization of the entire system.

A. LEAF DETECTION PERFORMANCE

The YOLO11 detector was trained to localize leaf regions from raw plant images. The fine-tuned YOLO11 achieved a precision of 97%, recall of 94%, and an F1-score of 93%. In addition, the detector obtained an mAP@0.5 of 91% and an mAP@0.5:0.95 of 88% on the validation set. These results show the effectiveness of the trained model in identifying individual leaves in different image conditions and complex backgrounds. After training, YOLO11 was integrated as the first stage of the pipeline to automatically crop leaf regions, which were then passed to the segmentation model. Samples from both PlantVillage and PlantDoc datasets, after leaf detection and cropping using YOLO11, are shown in Fig. 8 with their prediction confidences. The model reliably cropped and isolated leaf regions from both the laboratory test set and the real-world test set, serving as input to the segmentation stage.

B. LEAF SEGMENTATION PERFORMANCE

To completely and precisely isolate the leaf from its background, three segmentation methods were tested, including Marker-Based Segmentation, U-Net Segmentation, and Segment Anything Model (SAM). A visual comparison of the segmentation results obtained from the different methods is presented in Fig. 6. The figure illustrates the qualitative performance of each method in isolating leaf regions, showing differences in boundary precision and robustness to background variations, leaf shape variation, and appearance of disease symptoms. The visual comparison among the three methods showed that SAM produced cleaner and more accurate leaf masks. Although the U-Net method achieved satisfactory segmentation during training, its performance was negative when tested on the PlantVillage test samples.

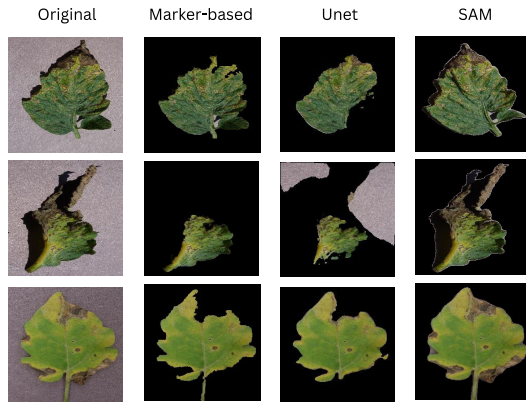


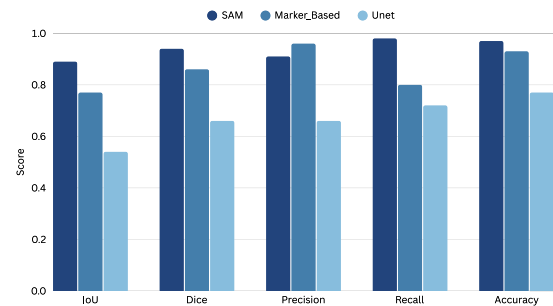
FIGURE 6. Comparison of segmentation outputs.

The ResNet-U-Net model often included background portions in its predicted masks, revealing overfitting to the uniform conditions of PlantVillage images and poor handling of natural variation. The marker-based method worked well for certain leaf classes, but was sensitive to changes in leaf color and texture, reducing effectiveness when disease symptoms altered leaf conditions. This is typical for most diseases listed earlier in Table 2, where infected leaves deviate significantly from healthy appearances and show changes in color and texture. SAM outperformed other alternatives, making it suitable for both laboratory and field images.

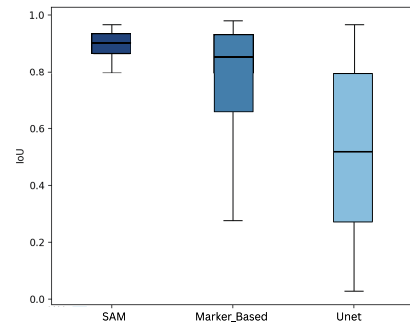
A manually annotated subset of 50 PlantVillage images served as ground truth for evaluation using Intersection-over-Union (IoU) and Dice Coefficient metrics. The evaluation results showed that SAM achieved the best segmentation performance, with IoU of 0.95, Dice of 0.96, Recall of 0.97, and Accuracy of 0.96, outperforming both the Marker-Based and U-Net methods. Although the Marker-Based method achieved slightly higher precision than SAM indicating that the method avoids labeling non-leaf regions, it misses some true leaf pixels. In contrast, SAM shows higher IoU, Dice, and Recall, showing that it captured the full leaf shape and boundaries more accurately. Since leaf segmentation tasks prioritize complete leaf extraction, SAM performs better overall. Its higher recall and overlap metrics indicate a more comprehensive and consistent segmentation of the leaf regions. Fig. 7a shows the comparison of segmentation metrics of the three methods and Fig. 7b shows the IoU distribution across segmentation samples. The distribution clearly shows that the SAM model delivers the most stable segmentation performance among the evaluated methods. Its IoU values are consistently high and tightly grouped, suggesting that the model performs reliably across different samples without large fluctuations. In contrast, Marker-Based Segmentation shows a wider spread, suggesting variable results across images, while U-Net indicates both lower and more scattered IoU values, reflecting limited

TABLE 4. Comparison of segmentation methods on the PlantVillage ground truth subset.

Method	IoU	Dice	Recall	Accuracy
Marker-Based	0.73	0.78	0.82	0.80
U-Net	0.88	0.90	0.92	0.91
SAM (Proposed)	0.95	0.96	0.97	0.96



(a) Comparison of mean segmentation metrics across methods.



(b) IoU distribution across segmentation samples.

FIGURE 7. Evaluation of segmentation performance. (a) IoU distribution of predicted masks and (b) comparison of mean segmentation metrics across models.

consistency. Moreover, SAM offers automatic and prompt-free segmentation, making it the best choice for accurate leaf extraction in the pipeline. Sample leaf images from both PlantVillage and PlantDoc datasets are shown in Fig. 8 with their SAM resulted images. The results, summarized in Table 4, show that SAM achieved the highest overall accuracy and generalization capability.

C. DISEASE CLASSIFICATION RESULTS (ResNet-50)

The classification training stage used the previously segmented leaf images as input to a ResNet-50. After 50 epochs, the model reached a training accuracy of 99.60% and a validation accuracy of 97.62% on the PlantVillage dataset. The learning curves, validation in dark blue and training in light blue, are shown in Fig. 9a and Fig. 9b, and the normalized confusion matrix for the PlantVillage validation set is provided in Fig. 10a. The decreasing validation loss shows that the model has learned from the training data and avoids overfitting. The training and validation curves

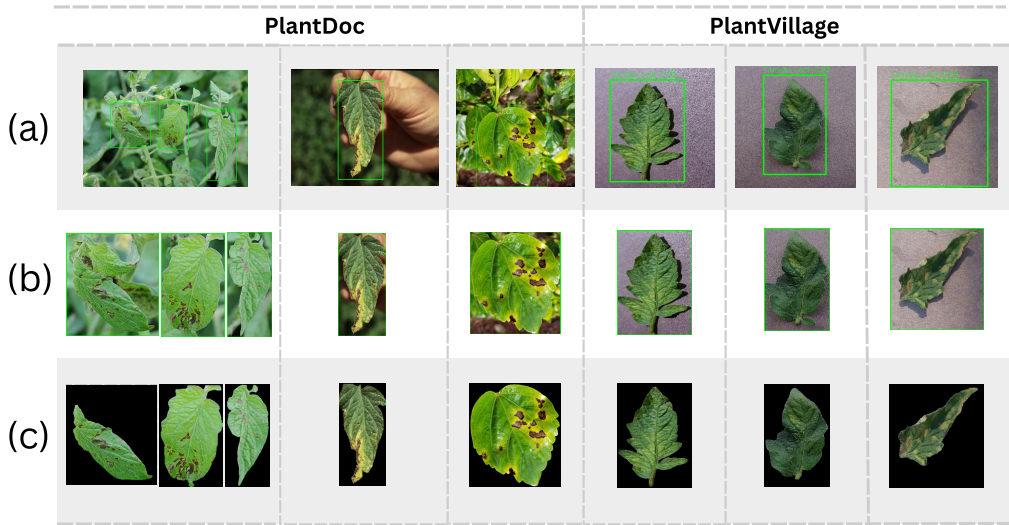


FIGURE 8. Leaf isolation results on samples from PlantVillage and plantDoc datasets (a) YOLO11 leaf detection (b) leaf cropping (c) SAM leaf segmentation.

remain close throughout, indicating good generalization and balanced learning. Accuracy improvements, both in training and in validation, indicate that the performance of the model has increased across epoches. Using the same hyperparameters and standard preprocessing stages but without previous leaf isolation, the flat ResNet-50 reached training accuracy of 99.70% and validation accuracy of 98.50%, indicating that it performs extremely well on the training dataset, slightly better than the trained pipelined model. The learning curves, validation in dark blue and training in light blue, are shown in Fig. 9c and Fig. 9d, and the normalized confusion matrix for the PlantVillage validation set is provided in Fig. 10b.

To evaluate the generalization ability of each trained model, we tested the performance of the proposed pipeline and the baseline model on both the laboratory dataset and the real-world dataset. The pipelined model achieved an accuracy of 96.76% and Macro-F1 of 0.9529 on the PlantVillage test set. While on the real-world PlantDoc set, it obtained an accuracy of 55.33% and Macro-F1 of 0.3656. This corresponds to an accuracy drop of 41.43 pp (42.82% relative) and a macro-F1 drop of 58.73 pp (61.63% relative). The normalized confusion matrix for the in the wild test is shown in Fig. 11a. The flat model achieved an accuracy of 98.21% and Macro-F1 of 0.9796 on PlantVillage, and accuracy of 22.79% and Macro-F1 of 0.1116 on PlantDoc. The corresponding drops were 75.42 pp (76.79% relative) in accuracy and 86.80 pp (88.61% relative) in macro-F1. Fig. 11b shows the confusion matrix of the experiment on PlantDoc. The previous results indicate that the flat ResNet showed high training and validation accuracies, but its performance degraded on the real-world PlantDoc test set, with dramatic drops in accuracy and macro-F1. In contrast, the pipelined model maintained a much better

generalization. Its results on PlantDoc exceeded those of the flat baseline, demonstrating the ability of the hierarchical design to mitigate domain shift. Table 5 summarizes the results of cross-domain evaluation for the two previously trained classifiers and shows the absolute accuracy drops using each of them.

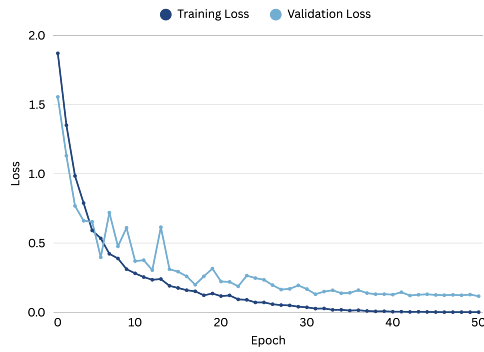
TABLE 5. Laboratory (PlantVillage) and real-world (PlantDoc) classification performance.

Model	Dataset	Accuracy (%)	Drop (pp)
Flat ResNet-50	PlantVillage	98.21	–
Flat ResNet-50	PlantDoc	22.79	75.42
Proposed Pipeline	PlantVillage	96.76	–
Proposed Pipeline	PlantDoc	55.33	41.43

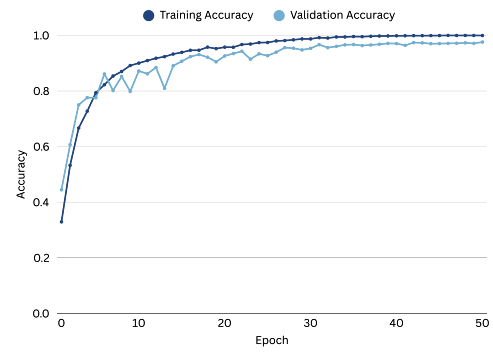
D. MODEL INTERPRETABILITY (LIME)

To improve transparency and reliability in the predicted diseases, the Local Interpretable Model-Agnostic Explanations (LIME) framework was integrated during inference. LIME highlights the regions most influential to the predictions of the model, producing heatmaps that visually create attention to diseased areas for expert evaluation. In addition, LIME is used to visually evaluate background dependency and its effect on generalization by comparing explanations from a flat ResNet classifier and from the pipelined model across the test datasets. Fig. 12 presents LIME maps from the flat classifier versus those from the pipelined model on the in-domain testing set. Using the flat model, the predictions depended on background features rather than solely on the target plant regions. However, after applying the proposed pipeline, the results showed more precise localization of disease regions.

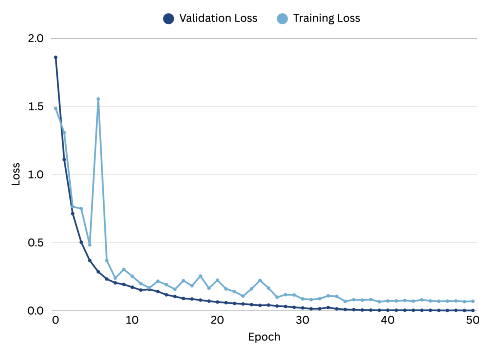
When moving to real-world conditions instead of controlled laboratory settings, the influence of background



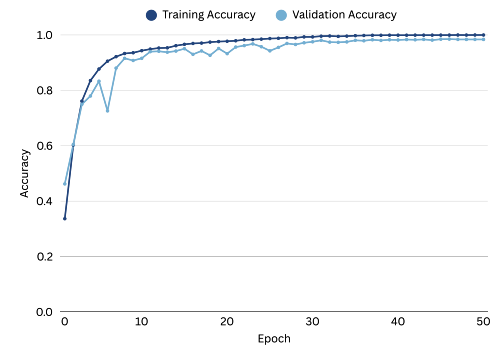
(a) Training and validation loss over 50 epochs of the proposed pipeline.



(b) Training and validation accuracy over 50 epochs of the proposed.

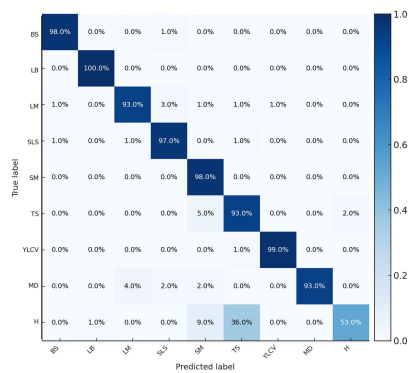


(c) Training and validation loss over 50 epochs of the flat ResNet.

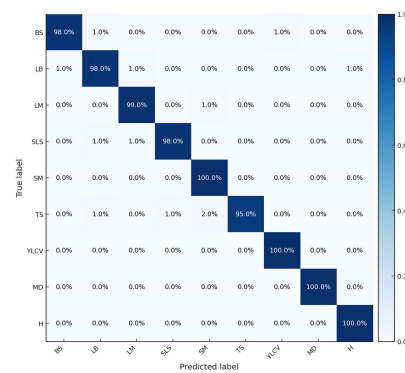


(d) Training and validation accuracy over 50 epochs of the flat ResNet training.

FIGURE 9. Training performance and validation evaluation of the proposed pipeline.(a–b) show the pipeline model accuracy and loss across 50 epochs.(c–d) show the flat ResNet accuracy and loss across 50 epochs.



(a) Normalized confusion matrix on the PlantVillage validation split (pipelined model).

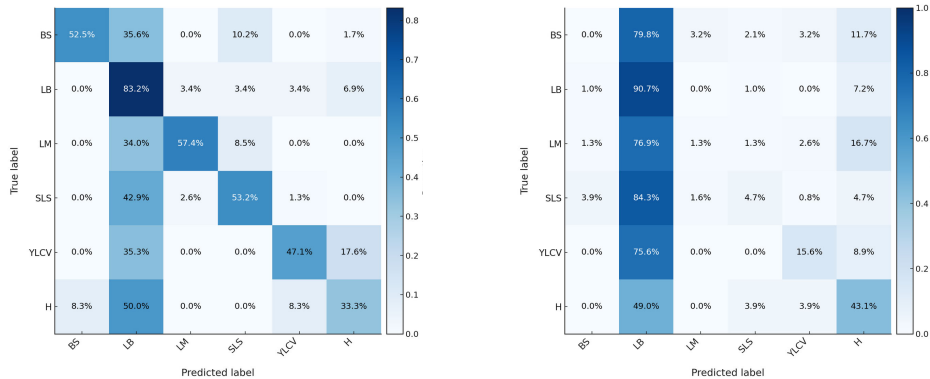


(b) Normalized confusion matrix on the PlantVillage validation split (flat model).

FIGURE 10. Normalized confusion matrices on the PlantVillage validation split.(a) Pipelined model. (b) Flat model.

features becomes even more pronounced. Fig. 13 shows a real-world sample image from the PlantDoc dataset with LIME highlighted regions before and after the pipeline. The captured image presented in Fig. 13a includes multiple leaves and stems from the background, making it more difficult for the model to focus on the actual diseased areas.

Using flat ResNet-50 without leaf isolation, LIME produced maps that focused on the background rather than the actual symptoms of the disease. After applying YOLO11 detection and cropping of leaf regions as shown in Fig. 13b, LIME attention was focused on the diseased regions on the leaf surface; however, the influence of the background remains



(a) Normalized confusion matrix on the PlantDoc testing split (pipelined model) (b) Normalized confusion matrix on the PlantDoc testing split (flatmodel).

FIGURE 11. Normalized confusion matrices on the PlantDoc test set.(a) Pipelined model. (b) Flat model.

TABLE 6. Comparison of model performance with existing studies on field (real-world) datasets.

Study	Dataset	Accuracy Drop (%)	Notes
[9]	PlantDoc	66–86	Large drop under real-world conditions.
[26]	Field dataset	27.81	Narrow field test set.
[24]	Synthetic (corruption-shifted)	15.4	Not fully real-world evaluation.
[25]	PlantDoc subset	26	Only 24 test samples used.
Proposed (Ours)	PlantDoc	41.43	Larger real-wrold subset, improved generalization.

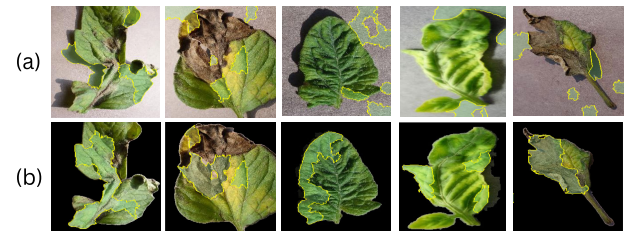


FIGURE 12. LIME-based interpretability results showing enhanced localization of diseased regions after the pipeline. (a) LIME results using the flat ResNet classifier, showing background attention; (b) LIME results from the full pipelined, showing attention aligned with disease symptoms.

visible. After SAM segmentation, the background effects on the classification process were completely ignored, focusing only on the region of interest and highlighting the accurate disease symptoms. This illustrates that the pipeline results in more reliable predictions in real-world images where conditions are complex and backgrounds are unexpected.

E. SUMMARY OF RESULTS

In general, the integrated pipeline combining YOLO11 for leaf detection, SAM for leaf segmentation, and ResNet-50 for classification achieved high training and validation accuracies and significantly narrowed the performance gap between controlled and field conditions during the testing process. These results indicate that the proposed pipeline improves generalization, robustness, and precise region

labeling compared to prior flat classifiers. The interpretability provided by the pipeline also makes it easier for users and domain experts to spot errors and mispredictions early and exclude irrelevant images. The overall findings highlight the potential of hierarchical, multi-stage vision pipelines for practical, field-ready crop disease diagnosis.

F. COMPARISON WITH EXISTING METHODS

In this section, we present a comparative analysis of various tomato leaf disease identification methods, focusing on generalization performance and comparing the proposed hierarchical pipeline with existing approaches and models.

Our proposed pipeline exhibited an accuracy drop of 41.43% when evaluated on the PlantDoc dataset, which is comparable to the 66–86 % drop reported by [9]. Although [26] achieved a 27.81% accuracy drop on a field dataset, the real-world dataset used contained only 665 images captured for this purpose. Furthermore, [24] reported a 15.4% accuracy drop; however, their evaluation was limited to corruption-based shifts rather than a completely real-world dataset. Although [25] achieved an accuracy drop of only 26% on PlantDoc, the subset used for the evaluation contained only 24 sample images. However, our evaluation contained a larger PlantDoc subset that represents tomato images in normal real-world scenarios.

By comparing our proposed hierarchical multi-stage pipeline to other approaches and models, we can see that the pipeline provides high training and validation accuracies, reduces the background effect on classification performance



(a) LIME results on an in-field sample image without the pipeline



(b) LIME results on an in-field sample image after leaf detection



(c) LIME results on an in-field sample image after segmentation and background removal

FIGURE 13. LIME results on an in-field sample image. (a) Irrelevant LIME highlights influenced by background noise, tomato fruit, and stems. (b) LIME results after leaf detection. (c) LIME results after leaf segmentation.

when tested on a comprehensive out-of-domain dataset, improves generalization by narrowing the gap between laboratory settings and real-world conditions, and enhances interpretability and explainability, enabling visualization to assess robustness and explain predictions for potential expert feedback. In addition, our work introduces an accurately labeled ground truth for a subset of the tomato PlantVillage dataset. This manually prepared ground truth is beneficial for evaluating different segmentation tools and approaches used for tomato leaf segmentation.

However, the remaining performance gap in real-world settings is still large, indicating that even the pipeline cannot fully overcome the biases of the dataset, lighting variations, or camera inconsistencies present in real agricultural images. Additionally, some PlantDoc samples may contain unclear or incorrect labels due to the challenges of annotating in-the-wild data, which may further influence the stability of quantitative performance. Detection and segmentation errors, such as detecting non-leaf objects, missing parts of a leaf, or including noise from the background, can propagate in the pipeline, affecting the classification performance. In addition, the trained model lacks the ability to identify multiple diseases in the same leaf image, which requires

using different classification architectures. However, from a practical perspective, this pipeline suggests a path for more robust and interpretable plant disease systems by structurally forcing the focus on leaf regions prior to disease classification. At the same time, the consistent evaluation protocol and strict separation from training process ensure that the observed performance reflects the behavior of the pipeline under realistic real-world conditions.

V. CONCLUSION

This study presented a hierarchical deep learning pipeline that integrates YOLO11 for leaf detection, the Segment Anything Model (SAM) for segmentation, and a ResNet-50 classifier enhanced with LIME-based interpretability for tomato leaf disease identification. The framework was designed to address the limitations of traditional classification models that operate on full plant images and fail to generalize to field conditions with complex backgrounds and lighting variations. The proposed pipeline ensures that the classification model focuses on the relevant areas of the image while removing background noise. This structure improved robustness and interpretability across both controlled (PlantVillage) and real-world (PlantDoc) datasets. Experimental results demonstrated that the proposed approach achieved an in-domain accuracy of 96.76% and an out-of-domain accuracy of 55.33% in testing datasets, outperforming baseline models trained without leaf isolation stages. SAM achieved the highest segmentation accuracy among the compared methods, with IoU = 0.95 and Dice = 0.96, confirming its ability to generalize under various environmental conditions. Furthermore, LIME-based visualization confirmed that the classifier's attention aligned with the actual symptomatic regions of the leaves, increasing the transparency and reliability of the predictions. Therefore, this pipeline establishes a strong foundation for the development of automated, interpretable, and field-ready plant disease diagnosis systems.

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